

HR Analytics Dashboard

Power BI Project Documentation

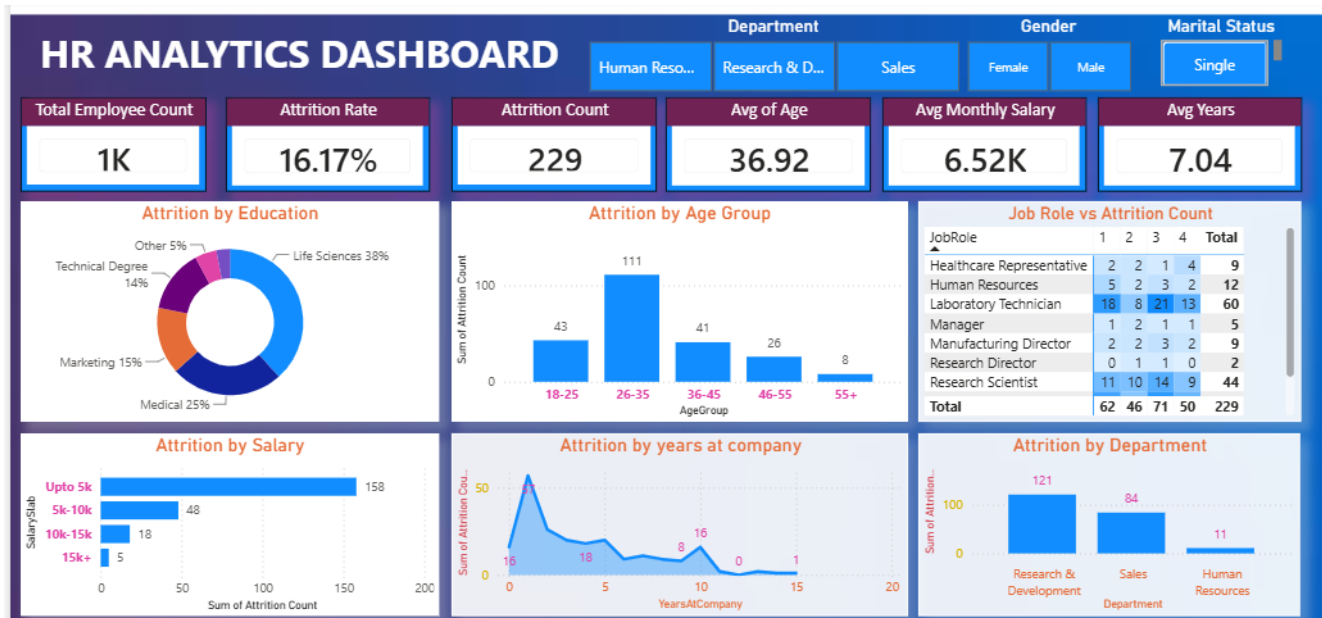


Figure 1: HR Analytics Dashboard

1. Project Overview

This project is an interactive HR Analytics Dashboard built using Power BI. It analyzes employee attrition and workforce trends across age groups, education fields, salary slabs, job roles, job satisfaction levels, years at company, and departments. The dashboard uses KPI cards, charts, a matrix, and interactive slicers to provide an overview of employee attrition.

2. Business Problem

Employee attrition is an important HR metric. The objective of this project is to understand employee turnover patterns and compare attrition across different employee and organizational dimensions.

Key business questions:

- What is the total employee count?
- What is the attrition rate and attrition count?
- Which age groups have higher attrition?
- How does attrition vary by salary slab?
- How does attrition vary across education fields?
- Which departments have higher employee exits?
- How is job satisfaction distributed across job roles?
- How does attrition vary with years at company?

3. Dataset

Source: Kaggle HR Analytics Dataset

<https://www.kaggle.com/api/v1/datasets/download/saadharoon27/hr-analytics-dataset>

Important fields used include:

- Employee Number
- Age
- Gender
- Department
- Education Field
- Job Role
- Job Satisfaction
- Monthly Income
- Years at Company
- Attrition

4. Tools and Technologies

- Power BI – dashboard and visualization
- Power Query – data cleaning and transformation
- DAX – KPI and analytical measures
- Excel – source dataset

5. Data Preparation

The data was prepared in Power BI using Power Query. Main preparation activities included:

- Checked data types
- Checked missing/null values
- Checked duplicate records
- Validated numerical and categorical fields
- Renamed fields where required
- Created Age Group and Salary Slab categories
- Validated the Attrition field
- Prepared the dataset for visualization

6. KPI Measures

6.1 Total Employees

Total Employees =
`DISTINCTCOUNT('HR Data'[EmployeeNumber])`

6.2 Attrition Count

Attrition Count = `CALCULATE([Total Employees], 'HR Data'[Attrition] = "Yes")`

6.3 Attrition Rate

Attrition Rate = `DIVIDE([Attrition Count], [Total Employees], 0)`

The Attrition Rate measure is formatted as a percentage.

6.4 Average Age

Average Age = AVERAGE('HR Data'[Age])

6.5 Average Monthly Salary

Average Monthly Salary = AVERAGE('HR Data'[MonthlyIncome])

6.6 Average Years at Company

Average Years = AVERAGE('HR Data'[YearsAtCompany])

KPI values are dynamic and change when slicers are applied.

7. Calculated Columns

7.1 Age Group

```
Age Group =  
SWITCH(  
    TRUE(),  
    'HR Data'[Age] < 25, "18-24",  
    'HR Data'[Age] < 35, "25-34",  
    'HR Data'[Age] < 45, "35-44",  
    'HR Data'[Age] < 55, "45-54",  
    "55+"  
)
```

7.2 Salary Slab

```
Salary Slab =  
SWITCH(  
    TRUE(),  
    'HR Data'[MonthlyIncome] < 5000, "Up to 5K",  
    'HR Data'[MonthlyIncome] < 10000, "5K-10K",  
    'HR Data'[MonthlyIncome] < 15000, "10K-15K",  
    "15K+"  
)
```

8. Dashboard Visualizations

KPI Cards: Total Employee Count, Attrition Rate, Attrition Count, Average Age, Average Monthly Salary, and Average Years.

Attrition by Education: Donut chart showing attrition distribution across education fields.

Attrition by Age Group: Clustered column chart showing attrition across age groups.

Attrition by Salary: Horizontal bar chart showing attrition across salary slabs.

Job Role vs Job Satisfaction Level: Matrix comparing job roles with job satisfaction levels.

Attrition by Years at Company: Line chart showing attrition count across years at company.

Attrition by Department: Column chart comparing employee attrition across departments.

9. Interactive Slicers

- Gender
- Department
- Marital Status

The Marital Status slicer is an additional filter included in the final dashboard. All KPI cards and visuals respond to the selected filter context.

10. Dashboard Design

- Bold dashboard title
- KPI cards positioned at the top
- Consistent fonts and formatting
- Data labels where useful
- Aligned visuals
- Clear spacing and borders
- Interactive slicers
- Consistent visual styling

11. Key Analysis Areas

- Overall employee attrition
- Age-wise attrition
- Salary-wise attrition
- Department-wise employee exits
- Education-wise attrition
- Job role and satisfaction patterns
- Attrition by years at company
- Filtered attrition by gender, department, and marital status

12. Project Structure

```
HR-Analytics-PowerBI-Dashboard/  
|  
├── README.md  
├── Dashboard/  
│   └── HR_Analytics_Dashboard.pbix  
├── Screenshots/  
│   └── HR_Analytics_Dashboard.png  
└── Documentation/  
    ├── Project_Documentation.docx  
    └── Project_Documentation.pdf
```

The source dataset is referenced through Kaggle rather than unnecessarily duplicating the dataset in the repository.

13. Project Outcome

This project demonstrates practical skills in:

- Power BI

- Power Query
- DAX
- Data Cleaning
- KPI Development
- Data Visualization
- Dashboard Design
- HR Analytics
- Business Analysis
- Interactive Reporting

The completed dashboard provides an interactive view of employee attrition and workforce trends and can be used as a portfolio project for Data Analyst roles.

Author

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Data Analytics Portfolio Project

Skills: Power BI | SQL | Excel | Python | DAX